Marcel Giovanni Munaba

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Education

Technische Universität München, M.Sc. in Computer Science — Apr 2026 – Present

- Specialization in Security and Machine Learning

Technische Universität München, B.Sc. in Computer Science — Oct 2022 – Mar 2026

- **GPA**: 2,3
- **Application field:** Mathematics
- **Elective Courses:** Fundamentals of AI, Introduction to Data Science and Statistical Thinking, Business Analytics and Machine Learning, Applied Regression, Patterns in Software Engineering

Hochschule Kaiserslautern, Internationales Studienkolleg — Mar 2022 – Jun 2022

- GPA: 1,5
- Preparatory program for international students without an Abitur-equivalent secondary-school certificate.

Note: In Germany, the range of passing grades is from 1.0 to 4.0, where 1.0 represents the highest mark.

Research

Quantifying Trade-Offs in the Blockchain Trilemma: Empirical Analysis in PoW-based Systems Towards Pareto-Optimal Configurations (Grade: 1,3), Bachelor's Thesis — February 2026

- **Research question**: Identify Pareto-optimal configurations for Proof-of-Work blockchains by measuring the blockchain trilemma subconcepts of decentralization, scalability, and security.
- **Methodology**: Deployed private Bitcoin Core networks with varied configuration parameters, then measured decentralization, scalability, and security.
- **Key Contributions**: Established baseline metric impacts, identified key correlations for Pareto efficiency, and developed a reusable framework for future trilemma analyses.

Experience

Working Student - Information Security, limango GmbH – Munich, Germany — December 2025 – Present

Engineering

- Designed and deployed an automated alert-notification pipeline connecting the SIEM to the team's collaboration platform.
- Develop automation scripts to tune firewall rules and reduce false positives, improving detection accuracy.

Operations

- Triage IDS alerts and analyze suspicious activity.
- Investigate and report on security incidents, such as malware detections, contributing to incident response and remediation efforts.

Working Student - IT Helpdesk, limango GmbH – Munich, Germany — June 2024 – December 2025

- Delivered desktop support and resolved 10+ technical issues weekly across Windows, macOS, and Linux, earning near-perfect user satisfaction.
- Managed 100+ user accounts, access permissions, security settings, hardware inventory lifecycle, and deployed and maintained company devices using a Mobile Device Management (MDM) tool to ensure compliance and streamline device provisioning.
- Supported onboarding and offboarding for up to 5 employees per month by configuring hardware, software, and access.

iPraktikum - iOS Developer, Interhyp Gruppe – Munich, Germany Oct 2024 – Feb 2025

- Partnered with a team of 8 developers in the *iPraktikum* course at TUM, led by the Chair of Applied Software Engineering in partnership with Apple.
- Engineered *HOMEvision*, a VisionOS app leveraging Google Gemini to extract natural, spoken descriptions of dream property characteristics and automatically translate them into smart search filters.

Projects

BenchBit

- A private Bitcoin testing framework, developed as part of the Bachelor's Thesis titled: *Quantifying Trade-Offs in the Blockchain Trilemma: Empirical Analysis in PoW-based Systems Towards Pareto-Optimal Configurations*.
- Tools Used: Python, Ansible

HOMEVision Project Page

- A VisionOS application developed in a team of 8, simplifying the home search process through personalized experiences. Using LLMs, it interprets user preferences expressed in natural language to suggest properties and provides immersive 3D visualizations of homes.
- Tools Used: Swift

Slash Survivors itch.io

- A top-down 2D pixel hack and slash game. Fight your way through endless waves of enemies and collect as much points as possible or work your way through objectives in normal mode.
- Tools Used: Godot

Note: More projects are available at my personal website.

Technologies

Programming Languages: Python, Java, C, Swift, JavaScript, Dart, R

Frameworks & Tools: scikit-learn, HTML/CSS, Node.js, SQL, Flutter, Godot, Ansible

Organizations

KMKI Bayern (Indonesian Catholic student organization in Bayern) - Food & Beverage Committee, Munich, Bavaria Aug 2023 – Jul 2024

- Managed and organized teams of cooks and kitchen staff across events with 50+ attendees.

SMAK 1 Cup Committee - Band Competition Division, Jakarta, Indonesia Oct 2020 – Dec 2020

- Assisted in finding, organizing and providing information to participants.

Languages

Indonesian (Native), English (C1), German (C1)